A Dodecahedral Transport Construction of the Arc-Transitive Graph AT4val[60,6], its Petersen Line-Graph Quotient, and the Thalean Structure Theorem

March 2026

Zenodo Preprint DOI:
https://doi.org/10.5281/zenodo.19043356

Abstract

We prove that local transport on the flag space of the combinatorial dodecahedron determines a connected 60-vertex tetravalent graph G_{60}, canonically identified with the arc-transitive graph AT4val$[60, 6]$ (House of Graphs: Graph52002).

The construction admits natural 2-fold quotients

$$G_{60} \to G_{30} \to G_{15},$$

where $G_{15} \cong L(\text{Petersen})$.

We show that the lifted transport induces a canonical $\mathbb{Z}_2$-valued cocycle on G_{15}, called the *Thalean cocycle*, which governs the signed 2-lift $G_{30} \to G_{15}$ and represents a nontrivial cohomology class.

From the transport system we construct a sector–edge incidence operator $M \in \mathbb{R}^{15\times 30}$ satisfying the polynomial identity

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I,$$

where A is the adjacency matrix of G_{15}.

After centering and normalization, this yields a spherical embedding of G_{15} in $\mathbb{R}^{14}$ with exactly three distinct inner products, determined by graph distance.

Thus the transport system decomposes into two complementary structures: a discrete $\mathbb{Z}_2$-valued cocycle capturing sheet holonomy, and a rigid spherical sector geometry arising from global incidence. Together these encode both the obstruction and the metric structure of the Petersen line-graph core.

1 Notation Summary

We summarize the primary objects used throughout the paper.

- G_{60}: chamber graph
- G_{30}: intermediate quotient
- G_{15}: deepest quotient, isomorphic to $L(\text{Petersen})$
- ε: transport invariant on chambers

- σ: induced edge sign function on G_{15}
- A: adjacency matrix of G_{15}
- M: sector incidence matrix

Cubic Thalean representation: the spherical embedding of $G_{15} \cong L(\text{Petersen})$ induced by the polynomial $p(x) = x^3 + 2x^2 + 2$ via the relation $MM^T = p(A)$

2 Introduction

The dodecahedron carries a rich combinatorial structure arising from its vertices, edges, faces, flags, and Petrie polygons. In this paper we show that local transport operations on the dodecahedral flag space give rise to a highly structured global object: a connected 60-vertex tetravalent graph G_{60}, identified as the arc-transitive graph AT4val$[60, 6]$.

Two natural involutive identifications produce quotient graphs

$$G_{60} \to G_{30} \to G_{15},$$

where the deepest quotient satisfies

$$G_{15} \cong L(\text{Petersen}).$$

Beyond this combinatorial identification, the transport system induces a canonical $\mathbb{Z}_2$-valued cocycle, which we call the *Thalean cocycle*. This cocycle governs a nontrivial signed 2-lift $G_{30} \to G_{15}$ and defines a nontrivial cohomology class.

From this cocycle we construct a sector–edge incidence operator M whose algebraic structure is governed by a polynomial identity in the adjacency operator,

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I,$$

where A is the adjacency matrix of G_{15}.

This identity defines a canonical polynomial representation of $L(\text{Petersen})$, which we call the *cubic Thalean representation*. After centering and normalization, this representation yields a three-angle spherical configuration in the 14-dimensional nontrivial adjacency module.

Thus, a single transport invariant — the Thalean cocycle — controls both the signed covering structure and a canonical geometric realization.

While the graph AT4val$[60, 6]$ is known, the transport-based construction, the Thalean cocycle, and the resulting cubic Thalean representation appear to be new.

3 Main Results

We summarize the core structure arising from dodecahedral flag transport.

Theorem 1 (Thalean Structure Theorem)**.** *Let G_{60} be the graph obtained from local transport on the dodecahedral flag space. Then the following hold.*

(i) ***Census identification.*** *The graph G_{60} is a connected tetravalent arc-transitive graph on* 60 *vertices, isomorphic to* AT4val$[60, 6]$ *(House of Graphs: Graph52002).*

(ii) ***Quotient structure.*** *There exist natural* 2*-fold quotients*

$$G_{60} \longrightarrow G_{30} \longrightarrow G_{15},$$

where

$$G_{15} \cong L(\text{Petersen}).$$

(iii) ***Transport cocycle.*** *The lifted transport induces a canonical* $\mathbb{Z}_2$*-valued cocycle*

$$\varepsilon : E(G_{15}) \to \mathbb{Z}_2,$$

which records whether transport across an edge preserves or swaps sheet in the double cover $G_{30} \to G_{15}$*. This cocycle is nontrivial in* $H^1(G_{15}, \mathbb{Z}_2)$*.*

(iv) ***Polynomial identity.*** *There exists a sector–edge incidence matrix* $M \in \mathbb{R}^{15\times 30}$ *such that*

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I,$$

where A *is the adjacency matrix of* G_{15}*.*

(v) ***Representation.*** *The map*

$$A \mapsto A^3 + 2A^2 + 2I$$

defines a canonical polynomial representation of the adjacency algebra of G_{15}*, induced by the transport system.*

(vi) ***Spherical sector geometry.*** *After centering and normalization, the rows of* M *define an embedding*

$$G_{15} \hookrightarrow \mathbb{R}^{14}$$

as a spherical configuration with exactly three distinct inner products, corresponding to graph distances $1, 2, 3$*.*

The transport system decomposes into two complementary structures:

- a discrete $\mathbb{Z}_2$-valued cocycle capturing sheet holonomy,
- a continuous spherical geometry arising from sector incidence.

Together these encode both the global obstruction and the metric structure induced by transport on the Petersen core.

4 Construction of the Chamber Graph G_{60}

We construct the chamber graph G_{60} from the flag structure of the combinatorial dodecahedron.

4.1 The dodecahedron

Let D be the combinatorial dodecahedron, with

$$|V(D)| = 20, \qquad |E(D)| = 30, \qquad |F(D)| = 12.$$

4.2 Flags

A *flag* of D is an incident triple

$$(v, e, f),$$

where v is a vertex, e an edge incident to v, and f a face incident to e.

There are 120 flags.

4.3 Local transport on flags

There are three natural involutive moves on flags:

- vertex move: change the vertex along the edge while keeping (e, f) fixed,
- edge move: change the edge within the face while keeping (v, f) fixed,
- face move: change the face along the edge while keeping (v, e) fixed.

These generate the adjacency structure on the flag graph.

4.4 Chamber states

Each edge is incident to four flags, which split into two opposite pairs. We define a *chamber state* to be such a pair.

Thus the 120 flags are partitioned into 60 chamber states.

4.5 Induced transport

The flag moves descend to well-defined operations on chamber states. Due to the identification of opposite flags, some flag moves coincide after descent.

A direct analysis shows that each chamber state admits exactly four distinct induced moves.

4.6 The chamber graph

The graph G_{60} is defined on the set of chamber states, with adjacency given by the induced transport moves.

Proposition 1. *The graph G_{60} is connected and 4-regular with 60 vertices.*

Proof. Each chamber state admits four distinct induced transport moves, giving degree 4. Connectivity follows from the connectedness of the flag graph and is verified computationally (Section 13). □

4.7 Quotient structure

The transport construction admits natural quotients

$$G_{60} \longrightarrow G_{30} \longrightarrow G_{15}.$$

The graph G_{15} has 15 vertices and 30 edges and will be identified with the line graph of the Petersen graph in the next section.

4.8 Notation

Throughout the paper:

- G_{60}, G_{30}, G_{15} denote the graphs defined above,
- edges are written as unordered pairs (u, v),
- all graphs are simple and undirected.

5 Quotient Structure

Starting from the 120 flags of the dodecahedron, we form 60 chamber states by identifying opposite configurations along edges. Local transport operations induce a connected tetravalent graph

$$G_{60},$$

with

$$|V| = 60, \quad |E| = 120, \quad \deg = 4.$$

Two natural involutive identifications yield quotient graphs

$$G_{60} \longrightarrow G_{30} \longrightarrow G_{15}.$$

Proposition 2 (Identification with the arc-transitive census)**.** *The graph G_{60} constructed via dodecahedral flag transport is isomorphic to the arc-transitive graph* AT4val$[60, 6]$. *In the House of Graphs database, this graph appears as Graph52002.*

Proof. The graph G_{60} is connected, 4-regular, and has 60 vertices and 120 edges. Computational verification shows that its automorphism group acts transitively on arcs and has order 480, and that the diameter is 6 with distance distribution $(1, 4, 8, 16, 24, 6, 1)$.

These invariants uniquely identify G_{60} within the census of tetravalent arc-transitive graphs, where it is recorded as AT4val$[60, 6]$. Cross-referencing with the House of Graphs database confirms the canonical identifier Graph52002.

□

6 The Petersen Line-Graph Core

In this section we identify the deepest quotient G_{15} with the line graph of the Petersen graph.

6.1 Line graphs

Let H be a graph. The line graph $L(H)$ has vertices corresponding to edges of H, with two vertices adjacent if and only if the corresponding edges of H share a common endpoint.

6.2 Main result

Theorem 2.

$$G_{15} \cong L(\text{Petersen}).$$

Proof. The graph G_{15} has 15 vertices and is 4-regular with 30 edges, matching the basic parameters of $L(\text{Petersen})$. Moreover, the adjacency relations agree with edge incidence in the Petersen graph. An explicit computation confirms that the two graphs are isomorphic. □

6.3 Interpretation

Under this identification, vertices of G_{15} correspond to edges of the Petersen graph, and adjacency corresponds to incidence in the Petersen graph; see, for example, [1].

7 Transport Invariant and Signed Lift

We now identify the binary transport invariant governing the lift

$$G_{30} \to G_{15}.$$

The key point is that the sign on a core edge is not assigned ad hoc: it is induced by the actual lifted transport and admits a canonical geometric interpretation on the dodecahedron.

7.1 The antipodal lift

The graph G_{30} is a 2-fold cover of G_{15}. Thus each vertex $x \in V(G_{15})$ has a two-point fiber

$$\pi^{-1}(x) = \{x^+, x^-\} \subset V(G_{30}).$$

For an edge $e = \{x, y\} \in E(G_{15})$, the adjacency upstairs connects the two fibers $\pi^{-1}(x)$ and $\pi^{-1}(y)$ in exactly one of two ways: either sheet is preserved, or sheet is exchanged. This binary distinction defines the transport parity of the edge.

7.2 Canonical dodecahedral model of the core

By the identification

$$G_{15} \cong L(\text{Petersen}),$$

we may regard the vertices of G_{15} as the 15 opposite-edge classes of the dodecahedron. Each vertex of G_{30} then corresponds to one of the two actual dodecahedral edges in the corresponding opposite-edge class.

Let e, f be two dodecahedral edges arising from adjacent vertices of G_{15}. Their relation is classified as follows:

- *local* if e and f share a vertex and lie on a common face;
- *nonlocal* if they neither share a vertex nor lie on a common face, and are not opposite edges.

Empirically, when one examines the two lifted edges in G_{30} above a single edge of G_{15}, only two patterns occur:

$$(LL) \qquad \text{or} \qquad (NN).$$

7.3 Definition of the Thalean cocycle

Definition 1 (Thalean cocycle)**.** *Let $e = \{x, y\} \in E(G_{15})$. Consider the two lifted edges in G_{30} connecting the fibers $\pi^{-1}(x)$ and $\pi^{-1}(y)$, and push them to the canonical opposite-edge model of the dodecahedron. Define*

$$\varepsilon(e) \in \mathbb{Z}_2$$

by

$$\varepsilon(e) = \begin{cases} 0, & \textit{if the lifted pattern is } (LL), \\ 1, & \textit{if the lifted pattern is } (NN). \end{cases}$$

Equivalently, $\varepsilon(e) = 0$ records trivial sheet holonomy across the edge, while $\varepsilon(e) = 1$ records nontrivial sheet exchange. This $\mathbb{Z}_2$-valued edge function is called the Thalean cocycle.

7.4 Cocycle theorem

Theorem 3 (Thalean cocycle theorem)**.** *The edge function*

$$\varepsilon : E(G_{15}) \to \mathbb{Z}_2$$

defined above has the following properties:

(i) It is canonically induced by the lifted transport $G_{30} \to G_{15}$.

(ii) It records whether transport across a core edge preserves or swaps sheet in the double cover.

(iii) Its cycle parity computes the sheet holonomy of lifted loops: for any cycle $C \subset G_{15}$,

$$\omega(C) := \sum_{e \in C} \varepsilon(e) \pmod 2$$

is trivial if and only if the lifted loop closes on the same sheet.

(iv) It defines a nontrivial cohomology class in $H^1(G_{15}, \mathbb{Z}_2)$.

Proof. The lift $G_{30} \to G_{15}$ provides a canonical two-sheet structure over each vertex of G_{15}. For each base edge, the induced adjacency between the two fibers is either sheet-preserving or sheet-swapping. This binary distinction is invariant under exchanging the two labels in a fiber, so it defines a well-defined element of $\mathbb{Z}_2$ on each edge.

Under the canonical identification of G_{15} with the graph of opposite edge-pairs of the dodecahedron, the two lifted realizations of each base edge are found to be simultaneously local or simultaneously nonlocal, yielding exactly the patterns (LL) and (NN). Thus the cocycle admits the stated geometric interpretation.

For a cycle C in G_{15}, the parity sum of edge values records whether the lifted walk in G_{30} returns to its original sheet or to the opposite sheet. Hence the cocycle parity computes the sheet holonomy of the loop. Computational verification shows that this class is nontrivial. □

7.5 Holonomy classes of pentagons

The cocycle does not create the pentagons of G_{15}; rather, it polarizes them into trivial and nontrivial holonomy classes. In particular, the parity sum

$$\omega(C) = \sum_{e \in C} \varepsilon(e) \pmod 2$$

distinguishes pentagons whose lifted transport closes trivially from those carrying nontrivial sheet-flip holonomy.

7.6 Signed-lift interpretation

If one passes from $\mathbb{Z}_2 = \{0, 1\}$ to $\{\pm 1\}$ by the rule

$$\sigma(e) = (-1)^{\varepsilon(e)},$$

then $\sigma(e) = +1$ corresponds to trivial holonomy and $\sigma(e) = -1$ to nontrivial holonomy. Thus the Thalean cocycle may be viewed equivalently as a signed edge function on G_{15} or as the $\mathbb{Z}_2$-voltage defining the double cover $G_{30} \to G_{15}$.

This cocycle is the fundamental transport invariant of the Petersen core and serves as the algebraic input for the sector incidence construction developed in the following sections.

8 Failure of Local Rules

We show that the transport invariant cannot be recovered from purely local combinatorial data on G_{15}.

8.1 Local indistinguishability

The graph G_{15} is 4-regular and edge-transitive. In particular, for any fixed radius r, all edges have isomorphic r-neighborhoods.

Thus no invariant depending only on finite-radius neighborhoods and respecting graph automorphisms can distinguish one edge from another.

8.2 Global variability of the cocycle

Despite this local symmetry, the Thalean cocycle

$$\varepsilon : E(G_{15}) \to \mathbb{Z}_2$$

takes both values 0 and 1.

Therefore the transport invariant distinguishes edges which are locally indistinguishable.

8.3 Failure of local determination

Theorem 4. *The Thalean cocycle is not determined by any rule depending only on finite-radius neighborhoods in G_{15} and invariant under graph automorphisms.*

Proof. Any such rule assigns values to edges based only on their r-neighborhoods up to isomorphism and is invariant under automorphisms.

Since G_{15} is edge-transitive, all edges have isomorphic r-neighborhoods. Hence any such rule must assign a constant value to all edges.

However, the cocycle ε takes both values 0 and 1 (see Section 13). Therefore it cannot arise from any such local rule. □

8.4 Geometric origin of nonlocality

The failure of locality is not accidental. The cocycle is defined by lifting edges to the double cover $G_{30} \to G_{15}$ and then resolving those lifted edges in the canonical dodecahedral model.

In that model, each edge of G_{15} corresponds to a pair of relations between dodecahedral edges, and the cocycle records whether these relations are both local or both nonlocal.

This information is invisible in G_{15} itself and only becomes accessible after passing to the lifted structure and its geometric realization.

8.5 Cohomological interpretation

The obstruction to locality is cohomological. The cocycle ε represents a nontrivial class in $H^1(G_{15}, \mathbb{Z}_2)$ and cannot be expressed as the coboundary of a vertex potential.

Equivalently, no vertex switching can trivialize the transport invariant.

8.6 Consequences

The transport invariant encodes genuinely global information: it is determined by lifted transport and geometric incidence in the dodecahedron, and cannot be reconstructed from local combinatorial data on the core graph alone.

Thus the Petersen line-graph core carries a hidden global structure revealed only through the Thalean lift.

9 Cohomological Classification

We interpret the transport cocycle as a cohomological object on G_{15}.

9.1 Cochains

Let $C^1(G_{15}, \mathbb{Z}_2)$ denote the space of $\mathbb{Z}_2$-valued functions on edges. Elements of C^1 assign a value in $\{0, 1\}$, equivalently a sign in $\{\pm 1\}$, to each edge.

Let $C^0(G_{15}, \mathbb{Z}_2)$ denote the space of $\mathbb{Z}_2$-valued functions on vertices. The coboundary map

$$\delta : C^0(G_{15}, \mathbb{Z}_2) \to C^1(G_{15}, \mathbb{Z}_2)$$

encodes vertex switching.

9.2 Cocycle from transport

The lifted transport $G_{30} \to G_{15}$ defines a function

$$\varepsilon : E(G_{15}) \to \mathbb{Z}_2,$$

where $\varepsilon(e)$ records whether transport across e preserves or swaps sheet.

For any cycle $C \subset G_{15}$, define the holonomy

$$\omega(C) = \sum_{e \in C} \varepsilon(e) \pmod 2.$$

By construction, $\omega(C)$ depends only on the cycle and computes the sheet parity of the lifted transport around that loop. Thus ε is naturally interpreted as a 1-cocycle.

9.3 Gauge equivalence

Switching at a vertex adds a coboundary. Hence two edge functions related by switching represent the same cohomology class. The transport cocycle therefore determines a well-defined element

$$[\varepsilon] \in H^1(G_{15}, \mathbb{Z}_2).$$

9.4 Dimension of the cohomology group

Because G_{15} is connected,

$$\dim H^1(G_{15}, \mathbb{Z}_2) = |E(G_{15})| - |V(G_{15})| + 1.$$

Since

$$|V(G_{15})| = 15, \qquad |E(G_{15})| = 30,$$

it follows that

$$\dim H^1(G_{15}, \mathbb{Z}_2) = 30 - 15 + 1 = 16.$$

Thus the full cohomology group is

$$H^1(G_{15}, \mathbb{Z}_2) \cong \mathbb{Z}_2^{16}.$$

The Thalean cocycle is therefore one distinguished nonzero class inside a 16-dimensional mod-2 cohomology space.

9.5 Nontriviality

Theorem 5. *The transport cocycle ε represents a nontrivial class in $H^1(G_{15}, \mathbb{Z}_2)$.*

Proof. A cocycle is trivial if and only if it is a coboundary, equivalently if all cycle holonomies vanish.

Explicit computation shows that there exist cycles in G_{15} with nontrivial holonomy. Therefore ε is not a coboundary and so defines a nontrivial cohomology class. □

9.6 Minimal support in the switching class

Within its switching class, the cocycle ε admits representatives with varying support sizes. Define the support

$$S = \{e \in E(G_{15}) : \varepsilon(e) = 1\}.$$

Proposition 3. *The minimal support size of ε over all switching-equivalent representatives is* 6.

Proof. An exhaustive search over all vertex switchings, with one vertex fixed to remove global symmetry, yields representatives with exactly 6 nonzero edges. No switching configuration produces fewer. □

Moreover, this minimal representative is not unique: the same exhaustive search yields 40 distinct supports of size 6.

Thus the number 6 measures the sparsest realization of the Thalean class, not the dimension of the ambient cohomology group.

9.7 Interpretation

The cohomology class $[\varepsilon]$ measures the obstruction to trivializing the double cover $G_{30} \to G_{15}$.

Equivalently, it records the existence of cycles whose lifted transport returns on the opposite sheet. Its minimal support gives a quantitative measure of how sparsely this obstruction can be represented after gauge reduction.

10 Geometric Interpretation

We interpret the transport construction geometrically, with particular attention to the relationship between the cocycle, the lifted transport, and the sector configuration on G_{15}.

10.1 Two kinds of structure on the core

The quotient graph $G_{15} \cong L(\text{Petersen})$ carries two distinct structures arising from transport:

- a $\mathbb{Z}_2$-valued cocycle ε on edges,
- a family of 15 transport sectors, one associated to each vertex.

The cocycle is sparse after switching reduction, while the sector system is highly distributed.

10.2 Geometric origin of the cocycle

Via the identification of G_{15} with the graph of opposite-edge classes of the dodecahedron, each vertex corresponds to a pair of antipodal edges, and each vertex of G_{30} selects one of the two.

An edge of G_{15} lifts to two relations between such edges. Each relation is either:

- *local*: the two edges share a vertex and lie on a common face,
- *nonlocal*: the two edges are neither incident nor cofacial and are not opposite.

Empirically, each edge of G_{15} yields either (LL) or (NN), and the cocycle records this dichotomy.

Thus the cocycle measures whether transport across an edge is geometrically local or nonlocal in the dodecahedral model. Equivalently, it records whether the lifted transport preserves or swaps sheet.

10.3 The cocycle support

Let

$$S = \{e \in E(G_{15}) : \varepsilon(e) = 1\}.$$

Before switching reduction, the cocycle support occupies a connected 14-edge subgraph. Its vertex-incidence profile is nonuniform: five vertices have incidence count 4, eight vertices have incidence count 1, and two vertices do not occur at all.

This shows that the cocycle selects a distinguished global region of the core which is invisible at the level of local graph structure.

10.4 The sector geometry

Independently, the transport sectors define a 15×30 incidence matrix M, whose centered row vectors produce a spherical configuration.

The centered inner products depend only on graph distance in G_{15} and take exactly three values:

$$2.4666666667, \qquad -1.5333333333, \qquad -2.5333333333,$$

corresponding to distances

$$1, \qquad 2, \qquad 3.$$

After normalization, the corresponding cosines are

$$0.3303571429, \qquad -0.2053571429, \qquad -0.3392857143.$$

Thus the sector system realizes a rigid three-angle spherical configuration in the nontrivial adjacency module.

10.5 Relation between cocycle and sectors

The cocycle and the sector geometry arise from the same transport system but capture different aspects of it.

- The cocycle records a binary obstruction: whether transport across an edge is local or nonlocal, equivalently whether it preserves or flips sheet in the lift.
- The sector system records higher-order overlap structure: how transport neighborhoods intersect globally across the core.

The sector identity

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I$$

shows that the spherical configuration depends only on graph distance, while the cocycle distinguishes edges which are otherwise identical at the level of G_{15}.

10.6 Nonlocality and emergence

The cocycle is not a function of the intrinsic graph structure of G_{15} alone. It arises only after lifting to G_{30} and resolving the lifted transport in the dodecahedral geometry.

By contrast, the sector configuration is already visible in the induced incidence algebra on G_{15}.

Thus the transport system exhibits two layers of structure:

- a hidden $\mathbb{Z}_2$-valued obstruction detected only through the lift,
- a continuous spherical geometry emerging from global incidence.

10.7 Relation to known graphs

The graph G_{60} is a connected tetravalent graph on 60 vertices with diameter 6, matching the standard identifier

$$\mathrm{AT4val}[60, 6].$$

The significance of the present construction is not merely that it recovers this graph, but that it equips it with:

- a canonical signed-lift structure over the Petersen core,
- a geometrically defined transport cocycle,
- and a rigid spherical sector geometry.

11 Sector Incidence Algebra and the Thalean Cocycle

Let $G \cong L(\text{Petersen})$ be the 15-vertex line graph of the Petersen graph, with adjacency matrix A.

11.1 Sector Incidence Operator

Let $M \in \mathbb{R}^{15\times 30}$ be the sector–edge incidence matrix arising from the Thalean transport construction. Each row corresponds to a vertex of G, and each column to an edge of G, with entries

$$M_{v,e} = \begin{cases} 1 & \text{if edge } e \text{ lies in the sector at } v, \\ 0 & \text{otherwise.} \end{cases}$$

Empirically,

$$\sum_e M_{v,e} = 14, \qquad \sum_v M_{v,e} = 7.$$

11.2 The Thalean Cocycle Identity

Theorem 6 (Thalean Cocycle Identity)**.** *Let A be the adjacency matrix of G. Then*

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I.$$

Proof. Let $p(x) = x^3 + 2x^2 + 2$. Since G has spectrum

$$\{-2^{(5)},\, -1^{(4)},\, 2^{(5)},\, 4^{(1)}\},$$

the operator $p(A)$ has eigenvalues

$$\{2^{(5)},\, 3^{(4)},\, 18^{(5)},\, 98^{(1)}\}.$$

Direct computation shows that MM^{T} has the same spectrum. Since both operators lie in the adjacency algebra of G, they coincide. □

11.3 Algebraic Interpretation

The operator M defines a canonical element of the Bose–Mesner algebra of G, with

$$MM^{\mathsf{T}} = p(A).$$

Thus, the transport system induces a polynomial representation of the adjacency algebra.

11.4 The Cubic Thalean Representation

Definition 2. *The map*

$$A \;\mapsto\; A^3 + 2A^2 + 2I$$

is called the cubic Thalean representation.

This representation is canonical: it is determined entirely by the transport-induced cocycle and the combinatorial structure of G.

11.5 Spectral Interpretation

If $Av = \lambda v$, then

$$MM^{\mathsf{T}} v = p(\lambda) v.$$

Thus, the representation acts by spectral transformation via $p(x)$.

12 Spherical Sector Geometry

12.1 Centering and Normalization

Let $\mathbf{1}$ denote the all-ones vector and define

$$M_c = M - \frac{14}{30}\mathbf{1}\mathbf{1}^T.$$

Then the rows of M_c lie in the 14-dimensional subspace orthogonal to $\mathbf{1}$. Normalize:

$$\widetilde{M}_v = \frac{(M_c)_v}{\|(M_c)_v\|}.$$

12.2 Three-Angle Structure

Theorem 7. *The vectors $\{\widetilde{M}_v\}$ form a spherical configuration in $\mathbb{R}^{14}$ with exactly three distinct inner products:*

$$\langle \widetilde{M}_u, \widetilde{M}_v \rangle \in \{\alpha, \beta, \gamma\}, \quad u \neq v,$$

where numerically

$$\alpha \approx -0.339286, \quad \beta \approx -0.205357, \quad \gamma \approx 0.330357.$$

Proof. The Gram matrix is obtained from MM^T after centering and normalization. Since $MM^\mathsf{T} = p(A)$ lies in the adjacency algebra and G has diameter 3, inner products depend only on graph distance. Direct computation yields exactly three values. □

12.3 Geometric Interpretation

The embedding $v \mapsto \widetilde{M}_v$ realizes G as a spherical three-angle configuration in $\mathbb{R}^{14}$.

The three inner products correspond exactly to graph distances $1, 2, 3$, showing that the geometry is completely determined by the adjacency structure.

12.4 Geometric Realization of the Representation

This embedding is the geometric realization of the cubic Thalean representation. The polynomial identity

$$MM^\mathsf{T} = p(A)$$

translates directly into metric relations in the embedding, making the geometry canonical.

13 Computational Verification

All constructions and claims relying on explicit enumeration are verified by deterministic computation.

13.1 Overview

A computational pipeline was implemented to construct the graphs G_{60}, G_{30}, and G_{15}, compute transport invariants, and verify all structural and algebraic claims stated in this paper.

All computations are based on explicit combinatorial data derived from the dodecahedral flag structure and are reproducible from the provided code and data files.

13.2 Graph construction and quotients

The graph G_{60} is constructed from chamber states of the dodecahedron. Two successive involutive identifications produce G_{30} and G_{15}.

The following properties are verified:

- G_{60}: 60 vertices, 120 edges, 4-regular,
- G_{30}: 30 vertices, 60 edges, 4-regular,
- G_{15}: 15 vertices, 30 edges, 4-regular,
- all graphs are connected,
- each quotient map is a 2-fold covering.

13.3 Identification of G_{15}

The graph G_{15} is verified to be isomorphic to the line graph of the Petersen graph by explicit graph isomorphism testing against a canonical construction derived from opposite edge-pairs of the dodecahedron.

13.4 Extraction of the transport cocycle

The Thalean cocycle is computed directly from the lifted adjacency in the covering $G_{30} \to G_{15}$.

For each edge $e = \{x, y\} \in E(G_{15})$, the two lifted edges between the fibers $\pi^{-1}(x)$ and $\pi^{-1}(y)$ are identified. These lifted edges are pushed to the canonical dodecahedral opposite-edge model, where each pair is classified as either:

- local (sharing a vertex and face), or
- nonlocal (neither incident nor cofacial and not opposite).

It is verified that only the patterns (LL) and (NN) occur, and the cocycle value is assigned accordingly:

$$\varepsilon(e) = \begin{cases} 0, & (LL), \\ 1, & (NN). \end{cases}$$

13.5 Edge sign distribution

The induced signed edge function on G_{15} is computed explicitly. In the representative gauge used for tabulation, the distribution is:

$$10 \text{ edges with } \varepsilon = 0, \qquad 20 \text{ edges with } \varepsilon = 1.$$

13.6 Holonomy verification

Cycle holonomy is verified directly.

For cycles $C \subset G_{15}$, the parity

$$\omega(C) = \sum_{e \in C} \varepsilon(e) \pmod 2$$

is computed and compared against lifted transport in G_{30}.

It is verified that:

- $\omega(C) = 0$ if and only if the lifted walk closes on the same sheet,
- $\omega(C) = 1$ if and only if the lifted walk returns on the opposite sheet.

In particular, pentagonal cycles split into trivial and nontrivial holonomy classes.

13.7 Cohomological classification

All vertex switchings (with one vertex fixed to remove global symmetry) are enumerated. The following are verified:

- the cocycle is not cohomologous to the trivial cocycle,
- no switching eliminates all nontrivial edges,
- the minimal support size is 6,
- exactly 40 distinct minimal supports of size 6 occur in the gauge-fixed search.

13.8 Sector incidence identity

The sector–edge incidence matrix M is constructed explicitly. The identity

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I$$

is verified by direct matrix computation.

13.9 Implementation notes

The transport cocycle and holonomy verification are implemented via explicit scripts operating on the graph data, including:

- construction of the antipodal quotients,
- identification with the canonical dodecahedral model,
- classification of lifted edge relations as local or nonlocal,
- cycle enumeration and holonomy testing.

13.10 Reproducibility

All data files, scripts, and outputs required to reproduce these computations are included in the repository. Details of the computational pipeline are provided in Appendices D and E.

14 Discussion

The construction produces two global structures:

- the *Thalean cocycle*, encoding a nontrivial $\mathbb{Z}_2$-obstruction to local transport;
- the *Thalean sector system*, encoding a rigid overlap structure governed by a polynomial in the adjacency operator.

Together, these show that local transport on the dodecahedral flag space generates both cohomological and geometric structure on the Petersen line-graph core.

This dual emergence suggests that transport systems on combinatorial geometries may systematically produce algebraic invariants and highly constrained spherical configurations.

Acknowledgements

The author acknowledges the use of AI-assisted tools in the development, structuring, and refinement of this manuscript.

15 Conclusion

We have constructed a transport system on the flag space of the dodecahedron, producing a 60-vertex chamber graph G_{60} with successive quotients

$$G_{60} \to G_{30} \to G_{15},$$

where $G_{15} \cong L(\text{Petersen})$.

This construction yields a nontrivial $\mathbb{Z}_2$-valued cocycle, realized as a signed 2-lift, whose minimal support has size 6 and which cannot be obtained from local combinatorial data.

From the transport structure we derive an algebraic identity relating the sector–edge incidence matrix to a polynomial in the adjacency matrix, and a corresponding spherical configuration with three distinct inner products.

These results demonstrate how global algebraic and geometric structure can emerge from local transport rules on a combinatorial object.

nputtables/signed$_edge_table$

A Explicit Edge Tables

This appendix records the edge structure of $G_{15} \cong L(\text{Petersen})$ together with the sign function σ defining the signed 2-lift.

We fix an ordering of the 30 edges of G_{15}, as listed below. All subsequent references to σ are understood with respect to this ordering.

The value $\sigma(e) \in \{\pm 1\}$ indicates whether the edge e lifts as a parallel edge $(+1)$ or a crossed edge (-1) in the double cover $G_{30} \to G_{15}$.

Edge	Sign	Type
$(0,1)$	-1	crossed
$(0,4)$	-1	crossed
$(0,5)$	$+1$	parallel
$(0,8)$	$+1$	parallel
$(1,2)$	-1	crossed
$(1,8)$	-1	crossed
$(1,11)$	$+1$	parallel
$(2,3)$	-1	crossed
$(2,11)$	-1	crossed
$(2,13)$	$+1$	parallel
$(3,4)$	-1	crossed
$(3,13)$	-1	crossed
$(3,14)$	$+1$	parallel
$(4,5)$	-1	crossed
$(4,14)$	-1	crossed
$(5,6)$	-1	crossed
$(5,12)$	$+1$	parallel
$(6,7)$	-1	crossed
$(6,12)$	-1	crossed
$(6,13)$	-1	crossed
$(7,8)$	-1	crossed
$(7,9)$	-1	crossed
$(7,13)$	$+1$	parallel
$(8,9)$	$+1$	parallel
$(9,10)$	-1	crossed
$(9,14)$	-1	crossed
$(10,11)$	-1	crossed
$(10,12)$	-1	crossed
$(10,14)$	$+1$	parallel
$(11,12)$	$+1$	parallel

B Minimal Cocycle Support

Let

$$S = \{e \in E(G_{15}) : \sigma(e) = -1\}.$$

The set S records the negative edges of the signing defining the signed 2-lift $G_{30} \to G_{15}$.

B.1 Gauge reduction

Two signings related by vertex switching define the same cohomology class. Explicitly, for a function $\tau : V(G_{15}) \to \{\pm 1\}$, the switched signing is given by

$$\sigma'(uv) = \tau(u)\,\sigma(uv)\,\tau(v).$$

Thus the support S is defined only up to switching equivalence.

B.2 Minimal support

We exhaustively search the switching orbit of σ, fixing one vertex to remove the global symmetry.

The computation shows that:

- the support can be reduced to size 6,
- no representative with fewer than 6 negative edges exists,
- there are 40 distinct supports of size 6 within the gauge-fixed search.

This establishes that the minimal weight of the cohomology class is 6.

B.3 Explicit representative

The table below records one representative of minimal support.

Edges in minimal support
$(0, 1)$
$(0, 8)$
$(2, 13)$
$(3, 13)$
$(10, 11)$
$(10, 12)$

B.4 Interpretation

Since the minimal support is nonzero, the cocycle defined by σ is nontrivial in $H^1(G_{15}; \mathbb{Z}_2)$.

The existence of multiple minimal representatives reflects the action of the switching group on the cohomology class.

C Dodecahedron Labeling

We fix a combinatorial model of the dodecahedron used throughout the paper.

C.1 Vertices and edges

Let the dodecahedron have 20 vertices and 30 edges. We label the vertices by a set $V = \{0, 1, \ldots, 19\}$.

Edges are unordered pairs (u, v) with $u, v \in V$. The full edge list is fixed once and for all and determines the combinatorial structure.

C.2 Faces

Each face is a 5-cycle in the edge graph. The 12 faces are recorded as cyclically ordered vertex lists

$$(f_1, f_2, f_3, f_4, f_5).$$

These face cycles determine the local adjacency relations used in the transport construction.

C.3 Flags

A flag is a triple (v, e, f) consisting of a vertex v, an incident edge e, and a face f containing e.

The dodecahedron has 120 flags in total.

C.4 Transport generators

The transport system is defined using local moves on flags:

- vertex move: change the vertex along a fixed edge,
- edge move: change the edge within a fixed face,
- face move: change the face across a fixed edge.

These generate the flag adjacency structure from which the chamber graph G_{60} is constructed.

C.5 Reproducibility

All computations in the paper use a fixed implementation of this labeling, consistent with the edge and face data used in the computational pipeline described in Appendix D.

D Computational Pipeline

This appendix describes the computational procedures used to verify the structural claims in the paper.

D.1 Graph construction

The graph G_{60} is constructed from the 120 flags of the dodecahedron by identifying opposite configurations to obtain 60 chamber states. Adjacency is generated by local transport moves.

The resulting graph is verified to be:

- connected,
- 4-regular,
- with 60 vertices and 120 edges.

D.2 Quotient computation

Successive quotients

$$G_{60} \to G_{30} \to G_{15}$$

are computed by identifying vertices under involutive symmetries.

The graph G_{15} is verified to be isomorphic to $L(\text{Petersen})$ by comparison of adjacency structure.

D.3 Signed lift and cocycle

A signing $\sigma : E(G_{15}) \to \{\pm 1\}$ is computed from the transport rules. The induced double cover is verified to reproduce G_{30}.

Switching equivalence is implemented via vertex sign assignments, and the minimal support is obtained by exhaustive search over the switching orbit.

D.4 Minimal support search

All 2^{14} switching configurations (fixing one vertex) are evaluated. For each configuration, the number of negative edges is recorded.

The minimum support size is found to be 6, and all minimal supports are enumerated.

D.5 Sector construction

The sector–edge incidence matrix M is constructed from the transport data. Matrix products are computed explicitly and verified to satisfy

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I,$$

where A is the adjacency matrix of G_{15}.

D.6 Implementation

All computations are carried out using Python scripts operating on explicit adjacency and incidence data. The computations are entirely deterministic and can be reproduced from the data and procedures described above.

E Reproducibility

All results in this paper are based on explicit finite computations on combinatorial data.

E.1 Data

The constructions use:

- a fixed labeling of the dodecahedron (Appendix C),
- an explicit edge list for G_{15} (Appendix A),
- a signing σ defining the cocycle (Appendix A).

E.2 Deterministic procedures

All computations are deterministic and involve:

- graph construction from flag transport,
- quotient identification,
- switching operations on signings,
- exhaustive enumeration over 2^{14} switching configurations,
- matrix computations on integer-valued data.

No randomness or probabilistic methods are used.

E.3 Verification steps

The main verifications can be reproduced by:

- constructing G_{60} from flag data and checking degree and connectivity,
- computing the quotients $G_{60} \to G_{30} \to G_{15}$,
- verifying $G_{15} \cong L(\text{Petersen})$,

- computing the signed lift from σ,
- enumerating switching configurations to confirm minimal support 6,
- computing the matrices A and M and checking

$$MM^{\mathsf{T}} = A^3 + 2A^2 + 2I.$$

E.4 Implementation

All computations are implemented in Python using explicit adjacency lists and incidence structures. The scripts operate directly on the data described in the appendices and require no external libraries beyond standard numerical and combinatorial functionality.

E.5 Scope

The results are fully reproducible from the data and procedures given in this paper. Any implementation following the same definitions will produce identical outputs.

References

[1] Chris Godsil and Gordon Royle. *Algebraic Graph Theory.* Springer, 2001.